Density Worksheet B

Questions

1. A lump of steel has a mass of 7600 kg and a volume of $1\,\text{m}^3$. What is the density of steel?
2. A container of water holds a volume of $2\,\text{m}^3$ of water; it weighs 2000 kg. What is the density of water?
3. A lump of copper has a mass of 18 400 kg and a volume of $2.3\,\text{m}^3$. What is the density of copper?
4. Brass has a density of $7500\,\text{kg/m}^3$. What is the mass if you have $2.1\,\text{m}^3$ of it?
5. Lead is very dense with a density of $13\,000\,\text{kg/m}^3$. If you have a sample with a mass of 2000 kg, what is the volume of this sample?
6. Gold has a mass of 45 kg. It has a density of $19\,300\,\text{kg/m}^3$. What is the volume of this gold?
7. A *cylinder* of brass $\left(\text{density} = 7500\,\text{kg/m}^3\right)$ has a radius of 0.01 m and a length of 0.05 m. What is its mass?
8. A *cube* of steel has a density of $7860\,\text{kg/m}^3$. If each side is 0.10 m long, what is the total mass of the steel cube?
9. A *cuboid* made of glass has the following dimensions: width 20 cm, height 20 cm, length 8 m. Calculate the density of this type of glass if the mass of this cuboid is 800 kg.
10. A steel *ball bearing* has a radius of 0.5 cm. Steel has a density of $7860\,\text{kg/m}^3$. Calculate the mass of the ball bearing.

Challenge Questions

11. A *concrete triangular prism* is 5 m long with a triangular cross-section of base 6 m. Its mass is 288 000 kg. Calculate the *height* of the prism.

    (Density of concrete: $2400\,\text{kg/m}^3$*.)*
12. A solid beryllium sphere has a mass of 3966 kg. Given that beryllium has a density of $1850\,\text{kg/m}^3$, calculate the radius of the sphere used in the particle accelerator.
13. A *hollow* spherical geode has an outer radius of 4 cm and a crust 0.5 cm thick made of limestone $\left(\text{density } 1760\,\text{kg/m}^3\right)$. Calculate the mass of just the limestone crust.

Useful formulae

Volume of a cube:	$V = a^3$
Volume of a cuboid:	$V = lwh$
Volume of a cylinder:	$V = \pi r^2 h$
Volume of a triangular prism:	$V = \frac{1}{2}bh\,l$
Volume of a sphere:	$V = \frac{4}{3}\pi r^3$
Density relationship:	$\rho = \dfrac{m}{V}$

Answer Key

1. $\rho = \dfrac{7600}{1} = 7600\,\text{kg/m}^3$
2. $\rho = \dfrac{2000}{2} = 1000\,\text{kg/m}^3$
3. $\rho = \dfrac{18400}{2.3} = 8000\,\text{kg/m}^3$
4. $m = \rho V = 7500 \times 2.1 = 15\,750\,\text{kg}$
5. $V = \dfrac{m}{\rho} = \dfrac{2000}{13000} = 0.154\,\text{m}^3$
6. $V = \dfrac{45}{19300} = 0.002\,33\,\text{m}^3$
7. $V = \pi r^2 h = \pi(0.01)^2(0.05) = 1.58 \times 10^{-5}\,\text{m}^3$
   $m = \rho V = 7500 \times 1.58 \times 10^{-5} = 0.118\,\text{kg}$
8. $V = a^3 = (0.10)^3 = 1.00 \times 10^{-3}\,\text{m}^3$
   $m = \rho V = 7860 \times 1.00 \times 10^{-3} = 7.86\,\text{kg}$
9. Convert the widths: $20\,\text{cm} = 0.20\,\text{m}$.
   $V = lwh = 8 \times 0.20 \times 0.20 = 0.32\,\text{m}^3$
   $\rho = \dfrac{800}{0.32} = 2500\,\text{kg/m}^3$
10. $V = \frac{4}{3}\pi r^3 = \frac{4}{3}\pi(0.005)^3 = 5.24 \times 10^{-7}\,\text{m}^3$
    $m = \rho V = 7860 \times 5.24 \times 10^{-7} = 0.004\,12\,\text{kg}$
11. (Triangular prism)
    $V = \dfrac{m}{\rho} = \dfrac{288000}{2400} = 120\,\text{m}^3$
    Cross-sectional area $A = \dfrac{V}{l} = \dfrac{120}{5} = 24\,\text{m}^2$
    $A = \frac{1}{2}bh \Rightarrow h = \dfrac{2A}{b} = \dfrac{2 \times 24}{6} = 8\,\text{m}$
12. (Sphere) $V = \dfrac{m}{\rho} = \dfrac{3966}{1850} = 2.14\,\text{m}^3$
    $r = \left(\dfrac{3V}{4\pi}\right)^{1/3} \approx 0.800\,\text{m}$
13. (Geode crust)
    $r_{\text{outer}} = 0.04\,\text{m},\ r_{\text{inner}} = 0.04 - 0.005 = 0.035\,\text{m}$
    $V_{\text{crust}} = \frac{4}{3}\pi(r_{\text{outer}}^3 - r_{\text{inner}}^3) = 8.86 \times 10^{-5}\,\text{m}^3$
    $m = \rho V = 1760 \times 8.86 \times 10^{-5} = 0.156\,\text{kg}$